PLUG PULLER · ONE-SIDED PULLER

Quick start, one-sided puller

The shortest path from a stuck plug to a printed one-sided puller that fits it and your hand: get the file, measure, fill in the four Customizer steps, print, assemble, use.

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Which tool this is

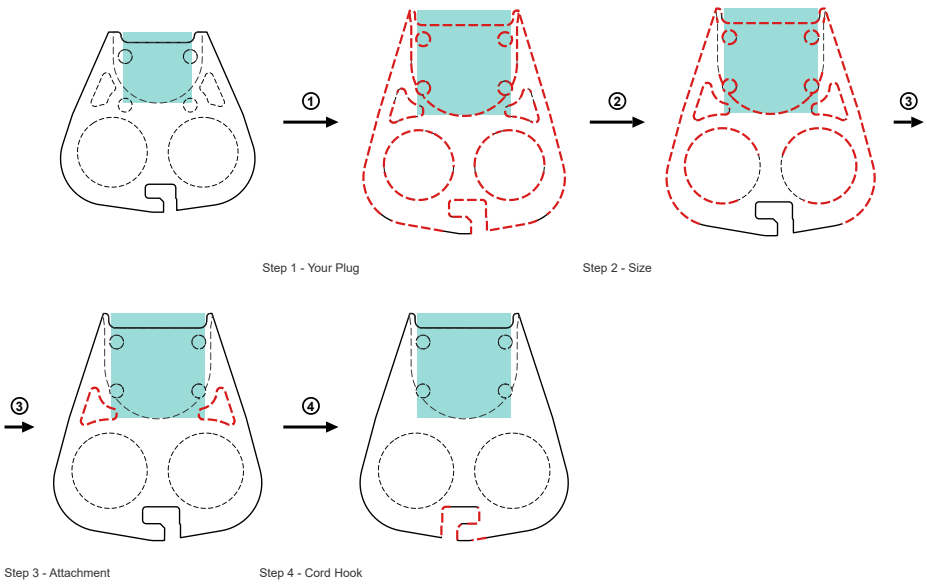
This guide is for the one-sided puller, src/Plug_Puller_Parametric.scad: the tool for a wall plug up to 24 mm thick, a pocket around the plug's back and sides with two finger holes below it, pulled with one hand. Measure your plug's thickness first. Thicker than 24 mm, or a plug you want held from both sides such as a USB-C tip or a round extension-cord plug, is the two-sided puller's job, and that tool has its own guide.

The Customizer's four steps are your plug, your size, the attachment and the hook's side. This quick start covers getting the file, measuring your plug and your hand or matching a card, each step's dials in a table, then printing, assembly and use.

> If red text appears beside the part in the preview, read it: it names the measurement to fix, and the part will not fit until it is gone.

How to read the pictures

- **Black:** the tool
- **Red dashed:** the edges this dial moved
- **Teal:** your plug
- ① the numbers match the key beside the picture



Five stages of the one-sided puller, left to right, the top row first, the plug end at the top; arrows numbered 1 to 4 show the steps, and red dashes mark the edges each step moved. Start, the defaults: the tool as the file opens, with a 25 mm wide plug in teal. Step 1 - Your Plug: plug length, plug width at both ends, plug thickness at both ends, cord thickness, outlet cover plate style; red on the body edge, finger holes, seat, wall notch, wing openings and zip-tie holes. Step 2 - Size: hand size, finger width, hand width; red on the body edge, finger holes, pocket, wall notch, wing openings and zip-tie holes. Step 3 - Attachment: attachment; the wing openings removed, drawn in red from the old outline. Step 4 - Cord Hook: hook side; red on the hook.

- Step 1: type your plug's numbers
- Step 2: pick your hand size
- Step 3: pick how it attaches
- Step 4: pick the hook's side

Get the file

The one-sided puller is the file `src/Plug_Puller_Parametric.scad`. There are two ways to open it and fill in its form: in your web browser with nothing to install, or in the free OpenSCAD program on your computer. Both show the same form, called the Customizer.

In your browser

The OpenSCAD Assistive Forge is a version of OpenSCAD that runs in your browser, on a computer or a phone, built for keyboard and screen reader use. Nothing is uploaded: your numbers stay on your device.

1. Open the one-sided puller in the Assistive Forge. On a first visit Forge asks which interface you want: choose **Assistive Forge**, then **Download & Continue**, and it downloads its engine once.
2. Forge offers to keep a copy of the project in your browser. Choose **Save My Copy**: the tool is then listed on Forge's main page whenever you come back, no link needed.
3. The form shows the four steps' dials first. The rest sit behind one button, **Show all parameters**.

If the link cannot load, Forge says why and offers **Try again**. You can also download the whole tool as one file, `dist/Plug_Puller_SingleFile.scad`, and open it from Forge's main page, under Open or start a project.

On your computer

OpenSCAD is the free program that turns your numbers into a printable file. You use one panel of it and never touch the code.

1. Download OpenSCAD from https://openscad.org/downloads.html: the **Development Snapshot** for your system. This project is tested with the snapshot of 2026-01-03; any recent snapshot works. The regular release works too, only slower.
2. Get the project: on its GitHub page press the green **Code** button, then **Download ZIP**, and unzip it anywhere. Or download only `dist/Plug_Puller_SingleFile.scad`, the whole tool in one file.
3. Open `src/Plug_Puller_Parametric.scad` (or the single file) in OpenSCAD: double-click it, or use **File ▸ Open**. A wall of code appears in an editor pane. Ignore it; you will not touch it.
4. Show the Customizer, the form you type into: in the **View** menu, make sure **Hide Customizer** is unchecked. In older versions it is **Window ▸ Customizer**.

The form's sections read top to bottom in the order you decide things: **Step 1 - Your Plug**, **Step 2 - Size**, **Step 3 - Attachment**, then **Step 4 - Cord Hook**. Everything below Step 4 is optional.

Measure your plug and your hand

Everything here is in mm: a US plug is about 25 mm wide, so a 1 on your paper means you measured in inches. You need a caliper or a ruler with mm marks, the plug in its outlet, and your own hand only if you pick Measure my hand. Print the measuring form at 100 % and fill it in as you go: the sections below are the form's rows, in the same order, and the card names (R1, C1, F1 / F2) are the cards of the printed measuring stencil, described under Match a card instead of measuring.

Form row	What to measure	How	Typical	Stencil card
2	**Plug length** measure_plug_length	With the plug in the outlet, ruler from the wall plate face to the plug's back face. On a laptop or charger plug, measure from the device's edge.	20 to 50 mm; example 38 mm	card R1
3	**Plug width at the prong end** measure_plug_width_prong_end	Caliper straight across the plastic body just behind the prongs, the wide way, parallel to the wall. Measure the body, not the metal prongs; if the head flares, take the widest part of that first stretch.	20 to 38 mm; example 34 mm	card R1
4	**Plug width at the cord end** measure_plug_width_cord_end	The same wide direction, but where the cord leaves the plug body. Skip any soft rubber strain relief and measure the last part of the hard body you would grip.	10 to 35 mm; example 13 mm	card R1
5	**Plug thickness at the prong end** measure_plug_thickness_prong_end	Caliper across the plug body's thin direction, usually top to bottom on a flat plug, just behind the prongs. A plug 24 mm thick or more needs the two-sided puller, so measure carefully.	12 to 30 mm; example 16 mm	card R1
6	**Plug thickness at the cord end** measure_plug_thickness_cord_end	The same thin direction, where the cord leaves the plug body, skipping the soft strain relief. On a round extension-cord plug this can be as fat as the prong end.	8 to 30 mm; example 9 mm	card R1
7	**Cord thickness** measure_cord_thickness	Caliper across the cord just behind the plug, on its thin side: flat cord the narrow way, round cord the diameter. With the stencil, the smallest C1 slot that slips over the cord is this number.	3 to 7 mm; example 5 mm	card C1
8	**Outlet cover plate style** measure_wall_plate_style	Look at the outlet cover plate: two small oval openings is Standard flat plate; one big rectangle per outlet is Rocker / Decora; a bigger, thicker plate is Oversized / Jumbo; no plate or a flush outlet is No plate / flush.	Choices: Standard flat plate, Rocker / Decora, Oversized / Jumbo, No plate / flush	none

Form row	What to measure	How	Typical	Stencil card
11	**Finger width** measure_finger_width	Caliper across the widest knuckle of your middle finger, the finger that goes into the pull hole. With the stencil, find the smallest F1 or F2 hole your finger passes through comfortably and subtract 5 from its number. No caliper? Take a ring that fits that finger snugly, measure the ring's inner diameter in mm and add 1.5 mm.	16 to 24 mm; example 22 mm	card F1 / F2
12	**Hand width** measure_hand_width	Caliper or ruler straight across the four knuckles of your flat hand, fingers together, no thumb. Measure at the widest point.	70 to 100 mm; example 88 mm	card R1
15	**Strap width** strap_width	Read the width on the strap's packaging. ONE-WRAP comes in 10, 13, 16, 20 and 25 mm.	10 to 25 mm; example 15 mm	none

> Sanity check: each plug width is a two-digit number, roughly 12 to 45, and the prong-end width is usually the bigger one; the finger width is roughly 14 to 32. A number like 1.3 is inches: measure again with the mm side.

When you measure for someone else, a relative or a client, measure their hand for the finger and hand rows and their outlet and plug for the plug rows. Doubtful between two values? Round up: a slightly roomy fit works, a tight one does not.

The measuring form

> The form is printed at true size on page 00, at the end of this guide; the paper stencil sheets follow it. Print the form at 100 % (actual size, never fit to page) and check its 50 mm bar with a ruler before you trust it. Fill in the blanks top to bottom as you measure, then type the numbers into the Customizer in the same order.

Match a card instead of measuring

The measuring stencil is a set of thin printed cards. Each has a two-letter name raised on its face, so you can find it by touch. Hold your plug in the plug cards: if it fills one card's openings, snug with no big gaps, that card names the preset to pick in Step 1, and you type no numbers at all.

Card	What it is	What it answers
P1	The flat 2-prong lamp plug (NEMA 1-15)	the preset `Flat 2-prong lamp plug - NEMA 1-15`
P2	The standard 3-prong plug (NEMA 5-15)	the preset `Standard 3-prong plug - NEMA 5-15`
P3	The heavy-duty extension cord plug (NEMA 5-15)	the two-sided puller's preset `Heavy-duty extension cord - NEMA 5-15`
P4	The wide 2-prong appliance plug (NEMA 1-15)	the one-sided puller's preset `Wide 2-prong appliance plug - NEMA 1-15`
R1	A ruler: raised mm ticks, numbers every 10 mm, and a notch every 10 mm you can count by touch	the plug's length, widths and thicknesses
C1	A cord gauge: open slots 3 to 9 mm wide along one edge, slid onto the cord from the side	the cord thickness: the smallest slot that slips over the cord
F1, F2	Finger holes 15 to 25 mm across (F1) and 26 to 32 mm (F2)	your finger width: the smallest hole your middle finger passes through comfortably, minus 5

Each P card has three openings: W for the plug's width, T for its thickness, and an open slot that slides onto the cord. If no card fits, your plug is between presets: measure it instead. The measured tool fits better than any preset.

Printing the cards. Print `stl/Measuring-Stencil/Visual/Measuring-Stencil_Visual_All-Cards.stl`: 1.2 mm thick cards, no supports, any rigid filament, packed onto sheets for a 200 by 200 mm bed. The tactile version, `stl/Measuring-Stencil/Tactile/`, prints every label as a raised character at ADA size and adds a Grade 2 braille title flap to every card. The full guide covers a smaller bed and folding the flaps.

No 3D printer yet? The last pages of the printed quick start and full guide are paper versions of the cards at true size: the plug outlines, a 100 mm ruler and the finger circles. Print them at 100 % (actual size, never fit to page) and check the 50 by 50 mm square with a ruler before you trust them. On their own they are `stencil-sheet.svg` and `stencil-sheet-2.svg`.

The four Customizer steps

Work the form top to bottom. Each step lists its dials: what each one does, then its default and the choices or the range it allows. The full guide shows a picture of what every dial moves.

Step 1 - Your Plug

Pick a plug preset, or leave it on Measure my plug and type your plug's numbers.

Dial	What it does	Default and choices
Plug preset plug_preset	Fills in all six plug numbers from a measured reference plug, so the pocket, the wall notch, the cord hook and the zip-tie and wing positions all take that plug's shape at once.	**Default:** Measure my plug **Choices:** Measure my plug, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15, Heavy-duty extension cord - NEMA 5-15, Wide 2-prong appliance plug - NEMA 1-15
Plug length measure_plug_length	Runs the pocket farther toward the finger holes and stretches the whole body to keep the finger holes and zip-tie rows clear of it.	**Default:** 25.5 mm **Range:** 12 to 85 mm, in steps of 0.5 mm
Plug width at the prong end measure_plug_width_prong_end	Widens or narrows the pocket, its seat and the wall notch at the plug end, and moves the zip-tie holes and wing openings with them.	**Default:** 25 mm **Range:** 8 to 38 mm, in steps of 0.5 mm
Plug width at the cord end measure_plug_width_cord_end	Tilts the pocket's side walls to match the plug's taper, and the zip-tie holes and wing openings lean in with them.	**Default:** 25 mm **Range:** 8 to 38 mm, in steps of 0.5 mm
Plug thickness at the prong end measure_plug_thickness_prong_end	Deepens or shallows the two pocket floors, the round seat and the plug recess, to suit the fatter of the two thickness numbers. The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.	**Default:** 20 mm **Range:** 4 to 40 mm, in steps of 0.5 mm
Plug thickness at the cord end measure_plug_thickness_cord_end	Deepens or shallows the two pocket floors, the round seat and the plug recess, when this end is the fatter end of the plug. The fatter of the two thickness numbers sets the floors, so lowering one alone changes nothing; 24 mm or more sends you to the two-sided puller.	**Default:** 20 mm **Range:** 4 to 40 mm, in steps of 0.5 mm
Cord thickness measure_cord_thickness	Widens the cord hook's slot and crossbar to fit the cord, and moves the finger holes and the whole body up to make room.	**Default:** 4 mm **Range:** 1.5 to 9 mm, in steps of 0.5 mm
Outlet cover plate style measure_wall_plate_style	Cuts the wall notch at the plug end deeper or shallower to suit the cover plate, and the zip-tie rows shift down with it.	**Default:** Standard flat plate **Choices:** Standard flat plate, Rocker / Decora, Oversized / Jumbo, No plate / flush

Dial	What it does	Default and choices
Show the see-through plug show_plug_preview	Shows or hides the see-through plug in the preview and changes nothing in the printed tool. Changes no shape: the see-through plug is a preview aid and is never exported.	**Default:** On **Choices:** On or off (a check box)

Step 2 - Size

Pick your hand size, or pick Measure my hand and type your hand's numbers.

Dial	What it does	Default and choices
Hand size size	Scales the whole body, both finger holes, the wing openings and the zip-tie grid to the chosen hand size.	**Default:** Medium **Choices:** Small, Medium, Large, Measure my hand, Custom
Finger width measure_finger_width	Widens both finger holes, spaces them farther apart and pushes them and the body up to keep the walls printable. Only acts when size is Measure my hand.	**Default:** 20 mm **Range:** 14 to 32 mm, in steps of 0.5 mm
Hand width measure_hand_width	Widens the whole body outline and thickens the slab in step with the hand. Only acts when size is Measure my hand.	**Default:** 85 mm **Range:** 60 to 110 mm, in steps of 1 mm

Step 3 - Attachment

Pick how the tool attaches.

Dial	What it does	Default and choices
Attachment attachment	Adds or removes the zip-tie holes and the wing or slot openings for the strap.	**Default:** Zip ties + Velcro **Choices:** Zip ties, Velcro strap, Zip ties + Velcro, None
Strap opening style velcro_style	Swaps the curved wing openings for a pair of plain rectangular slots that lean along the body's sides.	**Default:** Wing **Choices:** Wing, Classic slot
Strap width strap_width	Sets the strap width the wing opening is checked against and changes no shape. Changes no shape: it only sets the width the W-14 check compares the wing opening against.	**Default:** 15 mm **Range:** 10 to 25 mm, in steps of 1 mm

Step 4 - Cord Hook

Pick the hook's side.

Dial	What it does	Default and choices
Hook side hook_hand	Mirrors the cord hook at the cord end so its catch faces the other side, and nothing else moves.	**Default:** Right **Choices:** Right, Left

Render, export and print

1. On the computer, press **F6** (or **Design** ▸ **Render**) and wait for the progress bar to finish; a render takes from a few seconds to a minute. In Forge, the preview draws on its own: wait for **Preview ready** under the picture.
2. Look for red text beside the part. If there is any, read it: it names the measurement to fix, and the part will not fit until it is gone.
3. Save the file: on the computer, **File** ▸ **Export** ▸ **Export as STL**; in Forge, **Export STL**. If you export without rendering, OpenSCAD asks you to render first: press F6 and export again. The quick preview (F5) looks complete but cannot be exported.

Load the STL into your slicer with these settings:

Setting	Value
Orientation	flat face down on the bed, the plug pocket facing up. No supports.
Layer height	0.2 mm
Walls	3 to 4: you pull hard on this part
Infill	25 to 35 percent
Material	PETG. PLA, ABS and ASA work too. Never a flexible filament: the tool has to stay stiff.

The picture shows nine one-sided pullers printed this way, in three sizes, all flat on the bed.

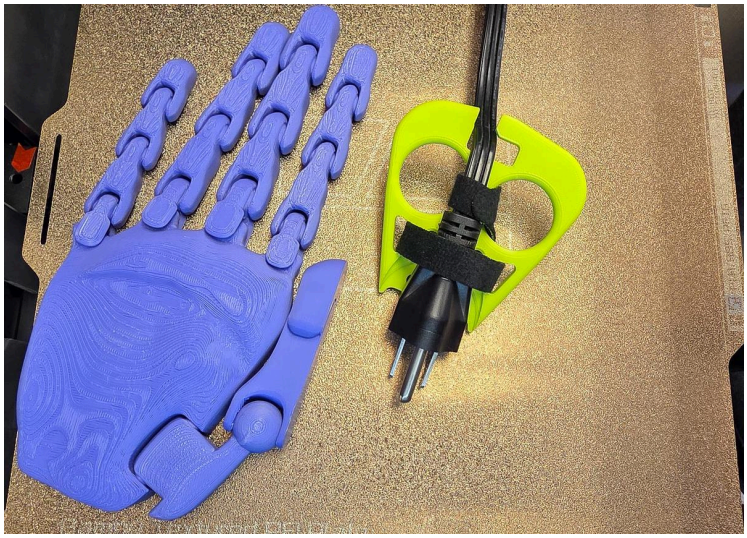


Nine one-sided pullers in three sizes printed flat on the printer bed, the pocket side up, no supports.

Assemble

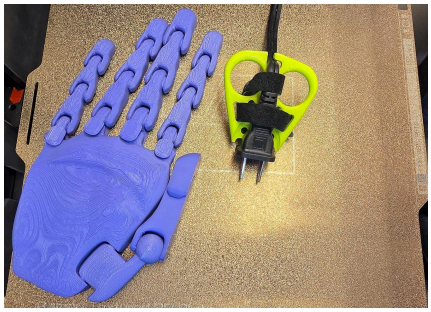
The tool goes onto the plug while the plug is in the outlet. You need two zip ties, or a strap; the sizes are in the bill of materials.

1. Seat the tool on the plugged-in plug: the plug sits in the pocket, and the notch at the plug end straddles the outlet cover.
2. Press the cord into the hook at the cord end so it cannot slip out.
3. Push the first zip tie down through one of the zip-tie holes beside the pocket.
4. Pass the tie around the plug body and up through the hole on the other side of the pocket.
5. Pull the tie tight and cut off the tail.
6. Fit the second zip tie through the other pair of holes the same way.
7. Instead of the ties, or as well: thread the strap through the two wing openings and around the plug, and close it. The picture shows a strap holding the tool on a three-prong plug.

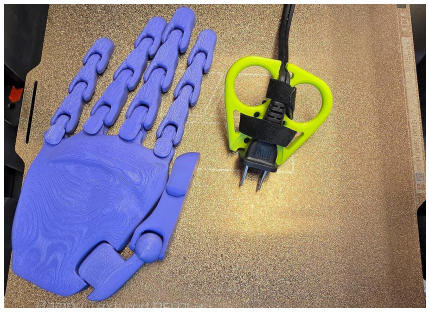


The one-sided puller at Medium size held on a black three-prong plug by a strap through its wing openings, on the printer bed beside a printed hand model for scale.

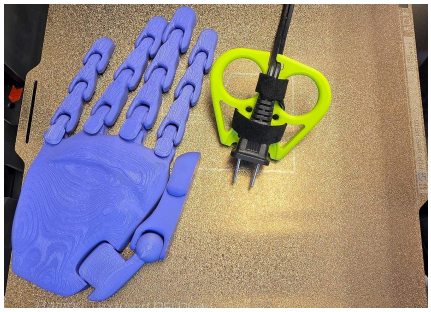
The tool can stay on the plug between uses. The pictures below show the finished one-sided puller strapped to a lamp plug at Small, Medium and Large, next to a printed hand for scale.



The one-sided puller at Small size strapped to a black two-prong lamp plug, on the printer bed beside a printed hand model for scale.



The one-sided puller at Medium size strapped to a black two-prong lamp plug, beside a printed hand model for scale.



The one-sided puller at Large size strapped to a black two-prong lamp plug, beside a printed hand model for scale.

Use it

The plug puller is for anyone who cannot grip a plug and pull it out of an outlet: arthritis, low grip strength, tremor, a small hand, one hand. Two finger holes take the pull, so your whole hand does the work instead of a fingertip pinch. The tool touches only the plug's sides and back, never the outlet.

1. Seat the tool on the plug while the plug is in the outlet: the plug in the pocket, the notch at the plug end over the outlet cover.
2. Press the cord into the hook at the cord end.
3. Put two fingers through the holes.
4. Pull straight back, away from the wall.

The tool can stay strapped to the plug between uses.

Safety

- Nothing goes between the plug face and the outlet cover: the tool grips only the plug's sides and back.
- Pull straight out. Never lever the tool sideways or up and down.
- Do not use the tool on a damaged cord, a cracked plug, or a plug that is warm to the touch.
- Keep your fingers away from the prongs as the plug comes out.
- The tool has no conductive parts, but it is still plastic near electricity: if it cracks, stop using it.

If it does not fit

Put the tool on the plug and pull once. The plug should sit flat in the pocket with the cord in the hook, and your two fingers should slide in and out of the holes without catching. If anything is tight, loose or awkward, the full guide names the one dial to change and by how much, decodes every red warning tag, and covers the advanced dials: the one-sided puller's full guide.

Plug Puller — Measuring form, one-sided puller

Measure in mm. Fill in the blanks top to bottom, then type them into the Customizer in the same order.

Print at 100 % — 50 mm

1 `plug_preset`
Plug preset
- (●) **Measure my plug**
- ( ) Flat 2-prong lamp plug - NEMA 1-15
- ( ) Standard 3-prong plug - NEMA 5-15
- ( ) Heavy-duty extension cord - NEMA 5-15
- ( ) Wide 2-prong appliance plug - NEMA 1-15

2 `measure_plug_length`
Plug length
default 25.5 mm → _____ mm card R1

3 `measure_plug_width_prong_end`
Plug width at the prong end
default 25 mm → _____ mm card R1

4 `measure_plug_width_cord_end`
Plug width at the cord end
default 25 mm → _____ mm card R1

5 `measure_plug_thickness_prong_end`
Plug thickness at the prong end
default 20 mm → _____ mm card R1

6 `measure_plug_thickness_cord_end`
Plug thickness at the cord end
default 20 mm → _____ mm card R1

7 `measure_cord_thickness`
Cord thickness
default 4 mm → _____ mm card C1

8 `measure_wall_plate_style`
Outlet cover plate style
- (●) **Standard flat plate**
- ( ) Rocker / Decora
- ( ) Oversized / Jumbo
- ( ) No plate / flush

9 `show_plug_preview`
Show the see-through plug
- (●) **leave on**

10 `size`
Hand size
- ( ) Small
- (●) **Medium**
- ( ) Large
- ( ) Measure my hand
- ( ) Custom

11 `measure_finger_width`
Finger width
default 20 mm → _____ mm card F1 / F2

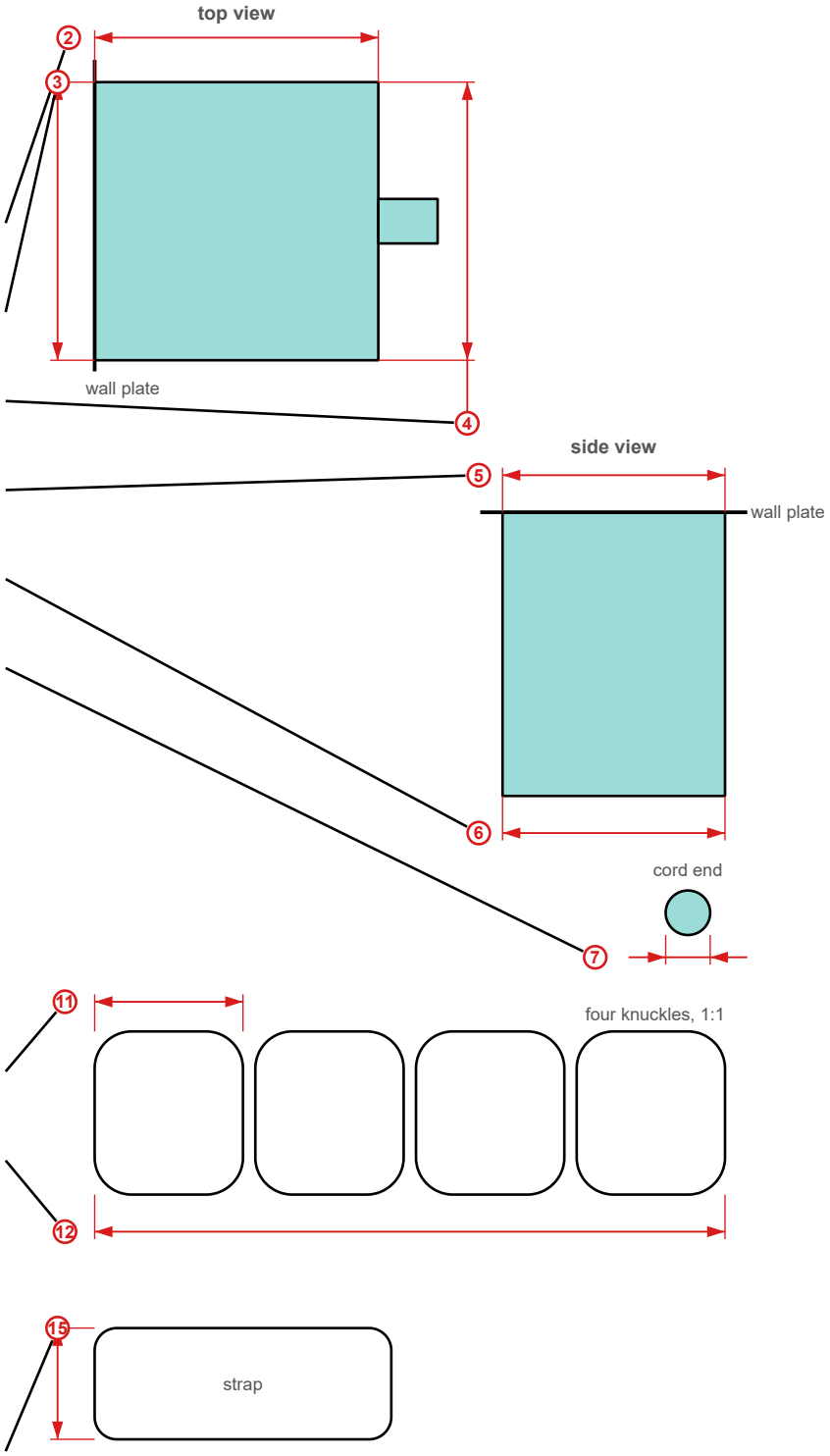
12 `measure_hand_width`
Hand width
default 85 mm → _____ mm card R1

13 `attachment`
Attachment
- ( ) Zip ties
- ( ) Velcro strap
- (●) **Zip ties + Velcro**
- ( ) None

14 `velcro_style`
Strap opening style
- (●) **Wing**
- ( ) Classic slot

15 `strap_width`
Strap width
default 15 mm → _____ mm

16 `hook_hand`
Hook side
- (●) **Right**
- ( ) Left



Plug Puller — Measuring Stencil Sheet

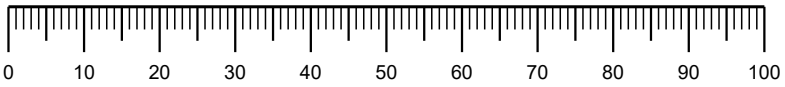
Print at 100% scale / “Actual size” — never “fit to page”. Check the calibration square before trusting anything on this sheet.

CALIBRATION
50 × 50 mm

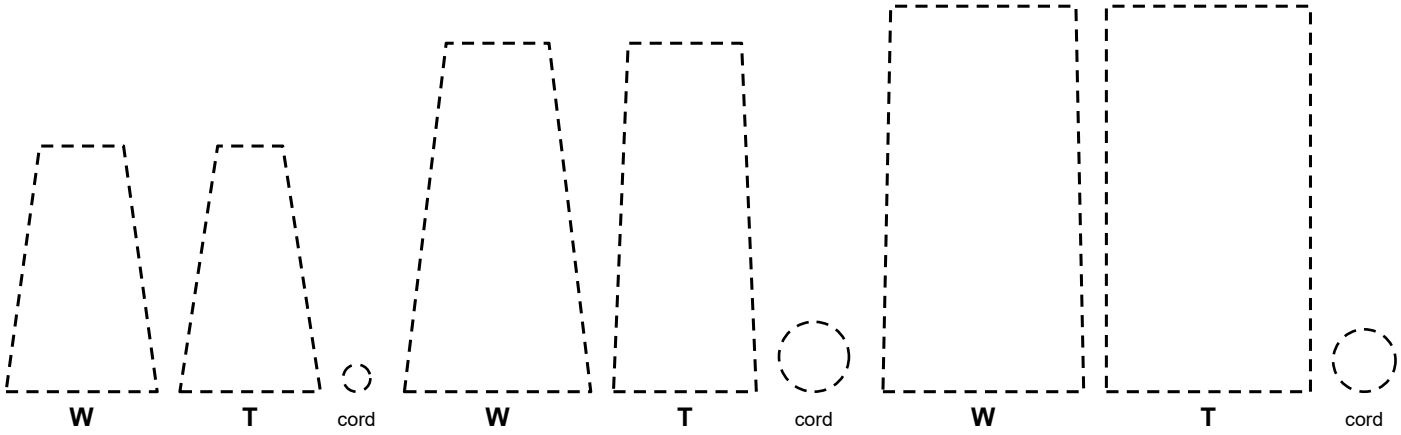
1. Measure this square with a ruler.
It must be exactly 50 × 50 mm — if not, your print was scaled: re-print at 100%.

Card legend: the quick start, under Match a card
Works on A4 and US Letter paper.

R1 — ruler (mm): plug length, widths, thicknesses, hand width

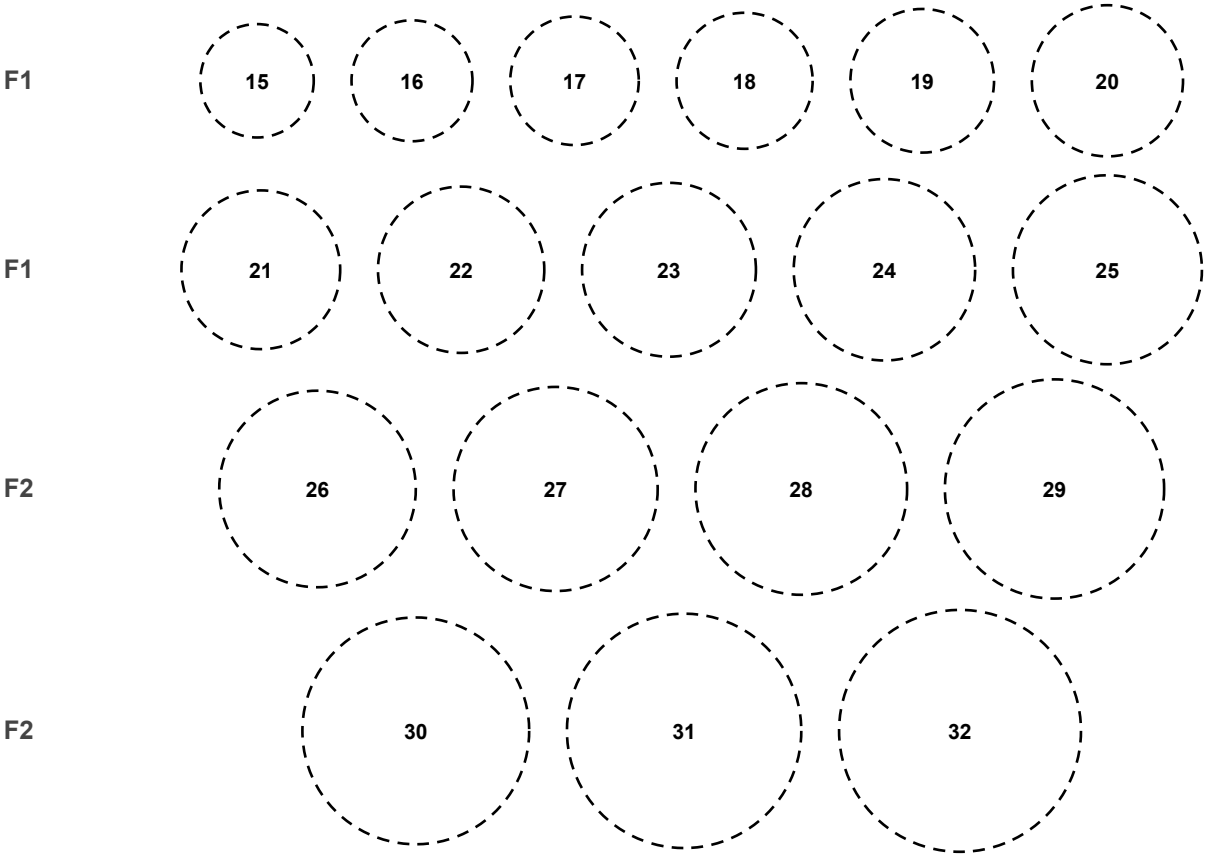


2. Plug silhouettes (1:1) — hold your plug against W (wide side) and T (thin side); if it matches, pick that preset in Step 1.



P1 Lamp 2-prong NEMA 1-15 **P2** Standard 3-prong NEMA 5-15 **P3** Heavy-duty cord NEMA 5-15

3. F1 / F2 — finger circles: smallest circle your middle finger passes through, minus 5 = finger width.



Plug Puller — Measuring Stencil Sheet

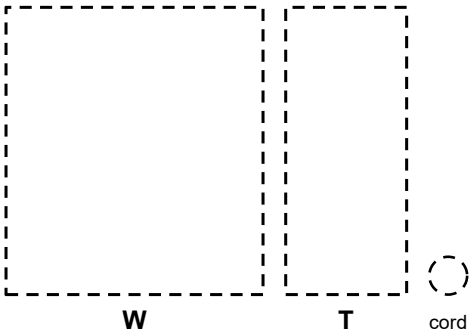
Print at 100% scale / “Actual size” — never “fit to page”. Check the calibration square before trusting anything on this sheet.

CALIBRATION
50 × 50 mm

1. Measure this square with a ruler.
It must be exactly 50 × 50 mm — if not, your print was scaled: re-print at 100%.

Card legend: the quick start, under Match a card
Works on A4 and US Letter paper.

2. Plug silhouettes (1:1), continued — hold your plug against W (wide side) and T (thin side); if it matches, pick that preset in Step 1.



P4 Wide 2-prong NEMA 1-15